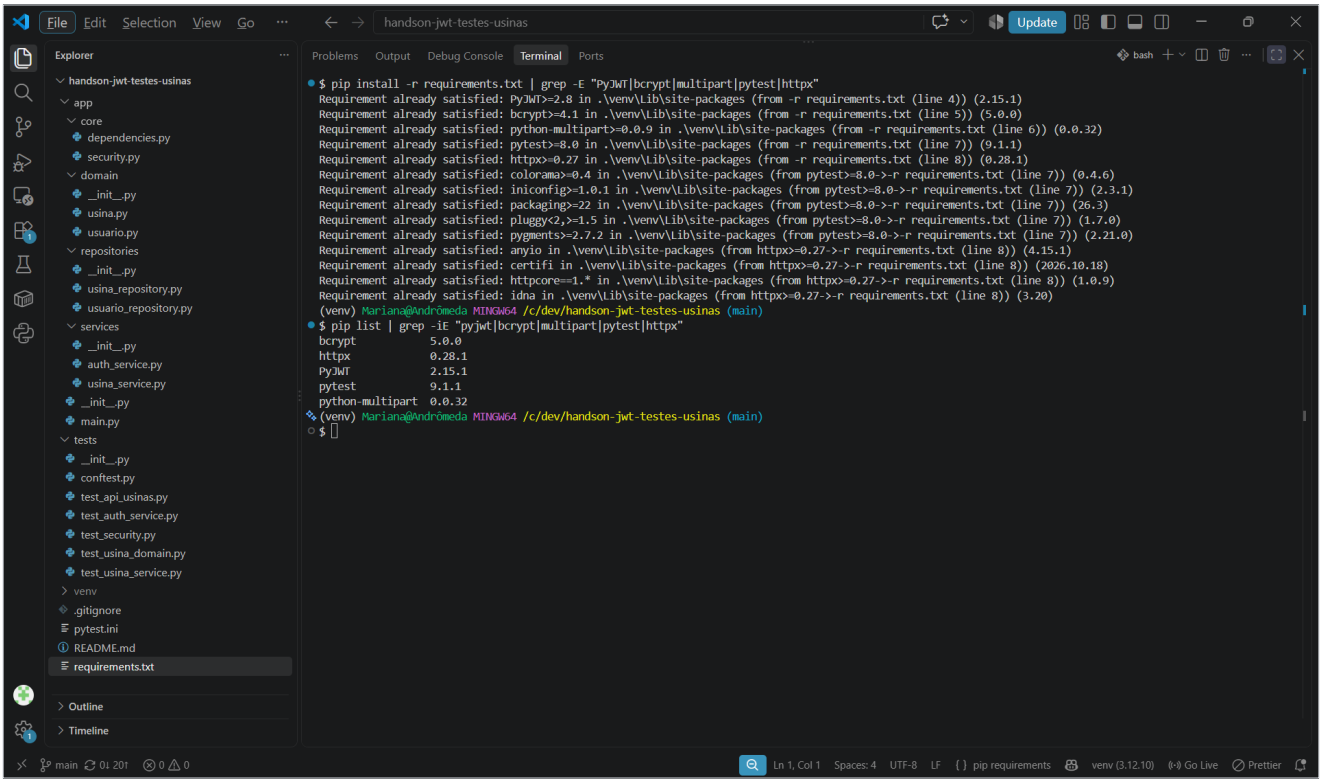
Handson: JWT e Testes Unitários no CRUD de Usinas

Passos 1 a 22

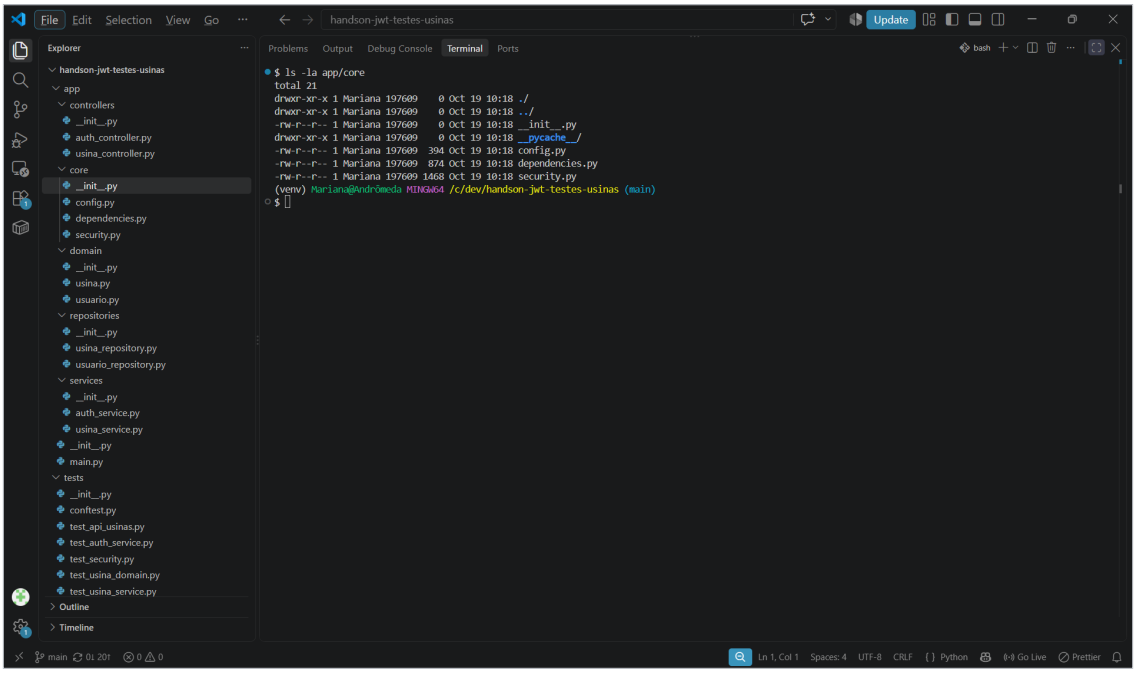
Disciplina: Projeto Ciência de Dados II, Bulbe Energia
Professor: Cristiano de Macedo Neto
Aluna: Mariana Maia
Data: 19/10/2026
Repositório: https://github.com/maia-mariana/handson-jwt-testes-usinas

Parte 1. Autenticação com JWT

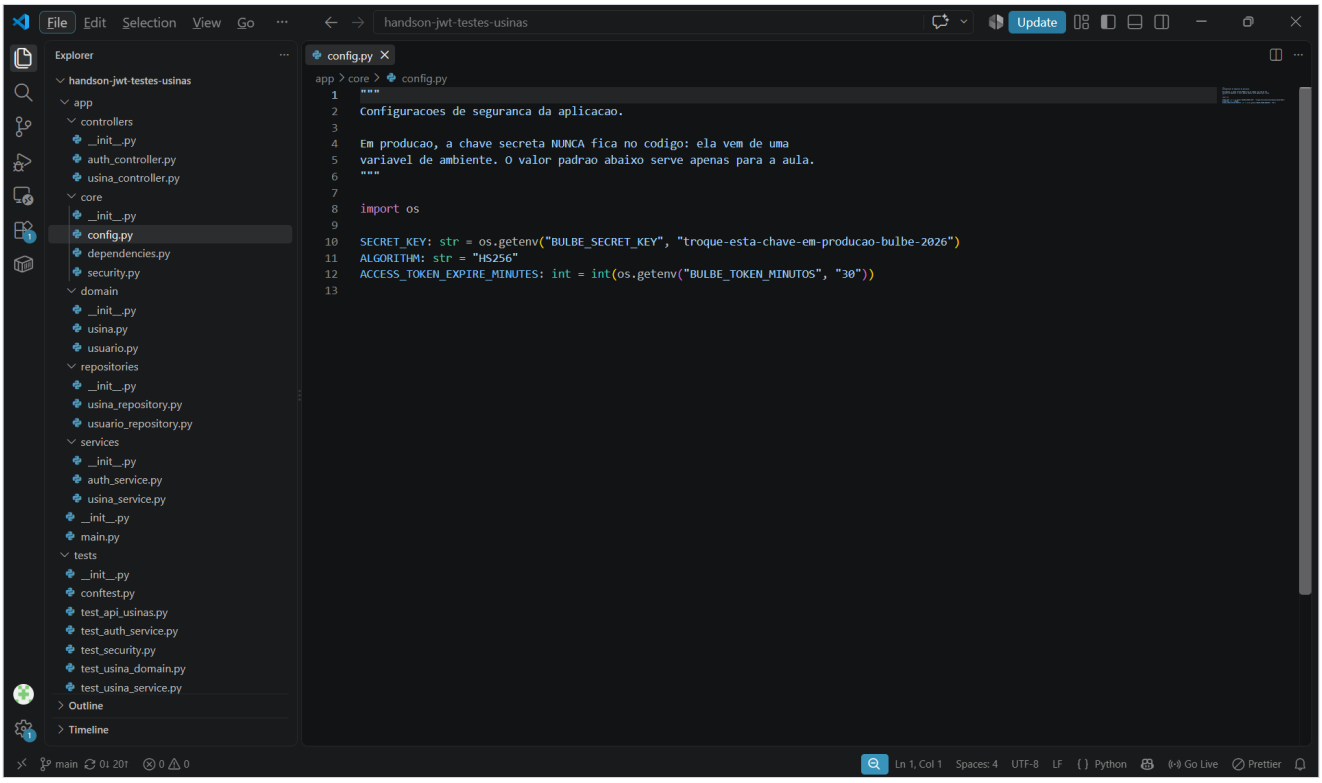
Passo 1. Atualizar as dependências



Passo 2. Criar a pasta core



Passo 3. Configuração: app/core/config.py



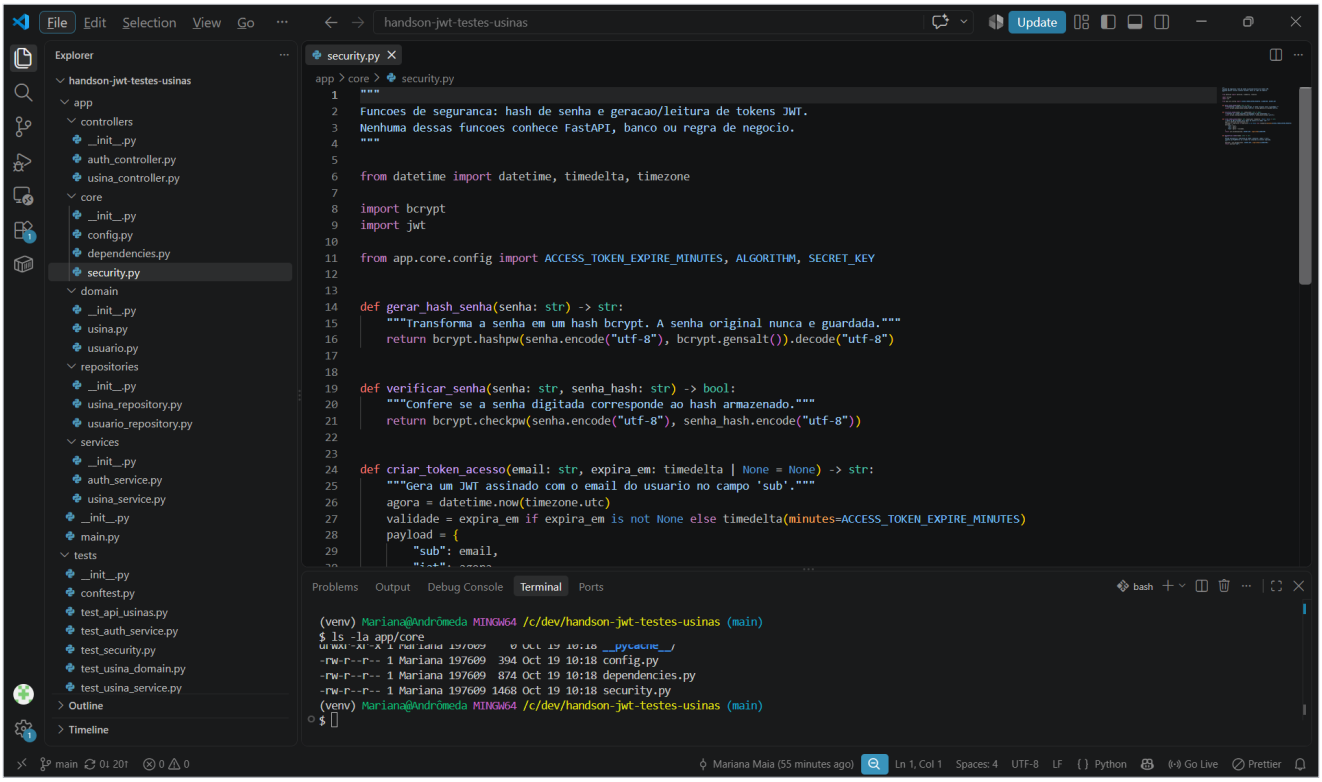
```
"""
Configuracoes de seguranca da aplicacao.

Em producao, a chave secreta NUNCA fica no codigo: ela vem de uma
variavel de ambiente. O valor padrao abaixo serve apenas para a aula.
"""

import os

SECRET_KEY: str = os.getenv("BULBE_SECRET_KEY", "troque-esta-chave-em-producao-bulbe-2026")
ALGORITHM: str = "HS256"
ACCESS_TOKEN_EXPIRE_MINUTES: int = int(os.getenv("BULBE_TOKEN_MINUTOS", "30"))
```

Passo 4. Segurança: app/core/security.py



```
"""
Funcoes de seguranca: hash de senha e geracao/leitura de tokens JWT.
Nenhuma dessas funcoes conhece FastAPI, banco ou regra de negocio.
"""

from datetime import datetime, timedelta, timezone

import bcrypt
import jwt

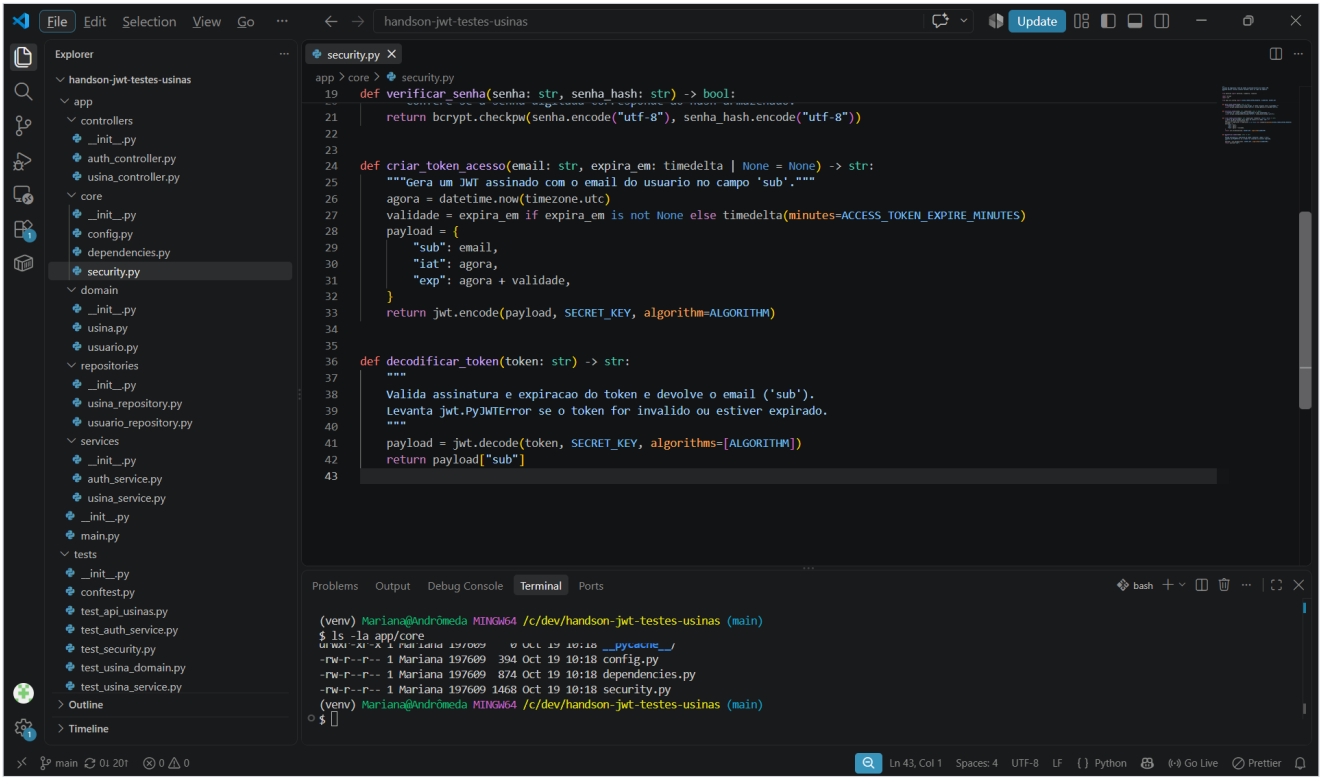
from app.core.config import ACCESS_TOKEN_EXPIRE_MINUTES, ALGORITHM, SECRET_KEY


def gerar_hash_senha(senha: str) -> str:
    """Transforma a senha em um hash bcrypt. A senha original nunca e guardada."""
    return bcrypt.hashpw(senha.encode("utf-8"), bcrypt.gensalt()).decode("utf-8")

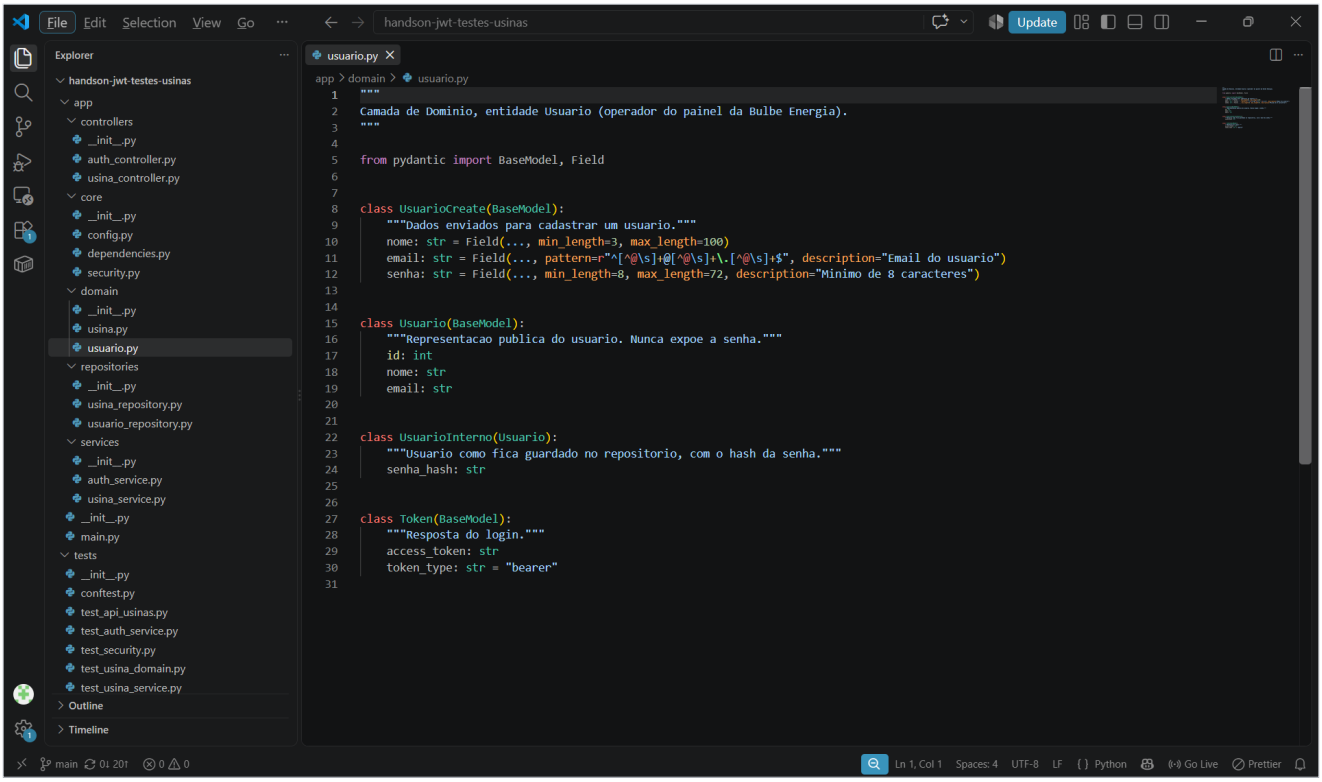

def verificar_senha(senha: str, senha_hash: str) -> bool:
    """Confere se a senha digitada corresponde ao hash armazenado."""
    return bcrypt.checkpw(senha.encode("utf-8"), senha_hash.encode("utf-8"))


def criar_token_acesso(email: str, expira_em: timedelta | None = None) -> str:
    """Gera um JWT assinado com o email do usuario no campo 'sub'."""
    agora = datetime.now(timezone.utc)
    validade = expira_em if expira_em is not None else timedelta(minutes=ACCESS_TOKEN_EXPIRE_MINUTES)
    payload = {
        "sub": email,
```

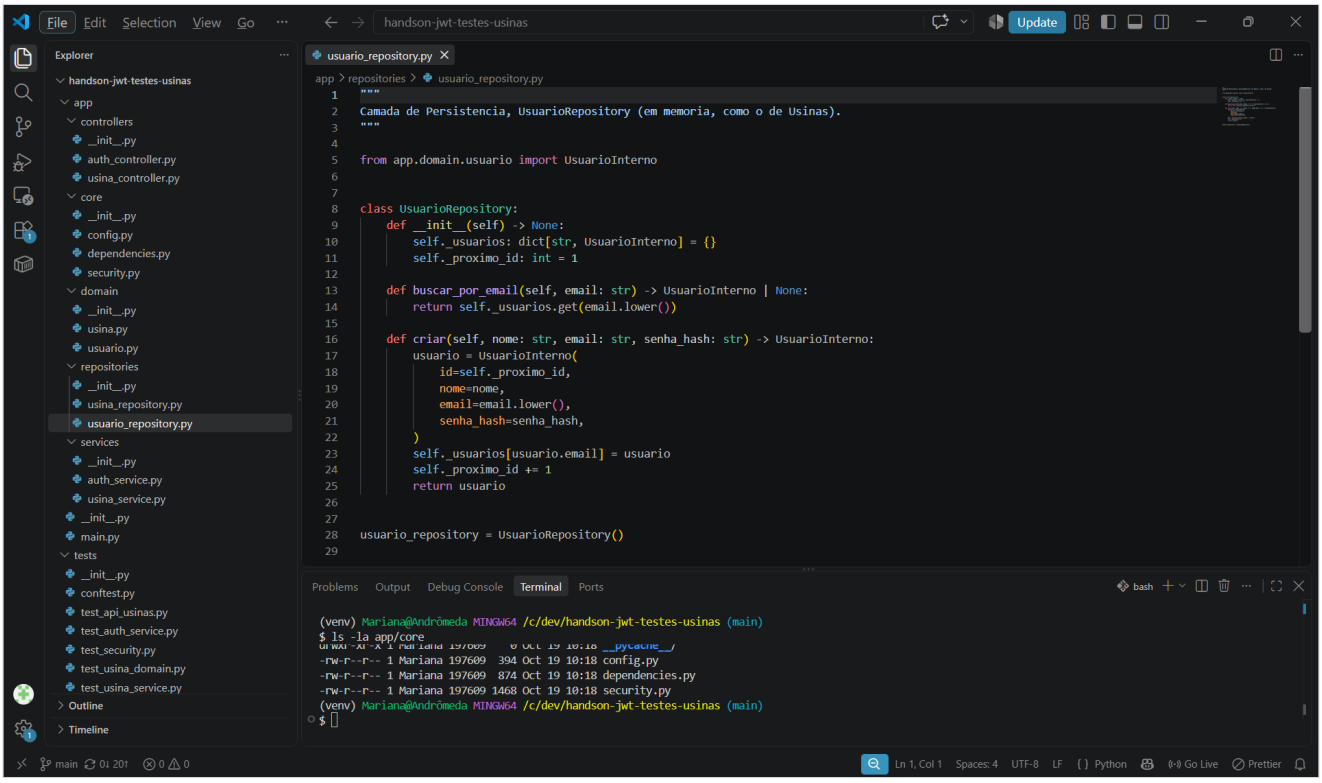
```
(venv) Mariana@Andrômeda MINGW64 /c/dev/handson-jwt-testes-usinas (main)
$ ls -la app/core
-rw-r--r-- 1 Mariana 197609  394 Oct 19 10:18 config.py
-rw-r--r-- 1 Mariana 197609  874 Oct 19 10:18 dependencies.py
-rw-r--r-- 1 Mariana 197609 1468 Oct 19 10:18 security.py
(venv) Mariana@Andrômeda MINGW64 /c/dev/handson-jwt-testes-usinas (main)
$
```



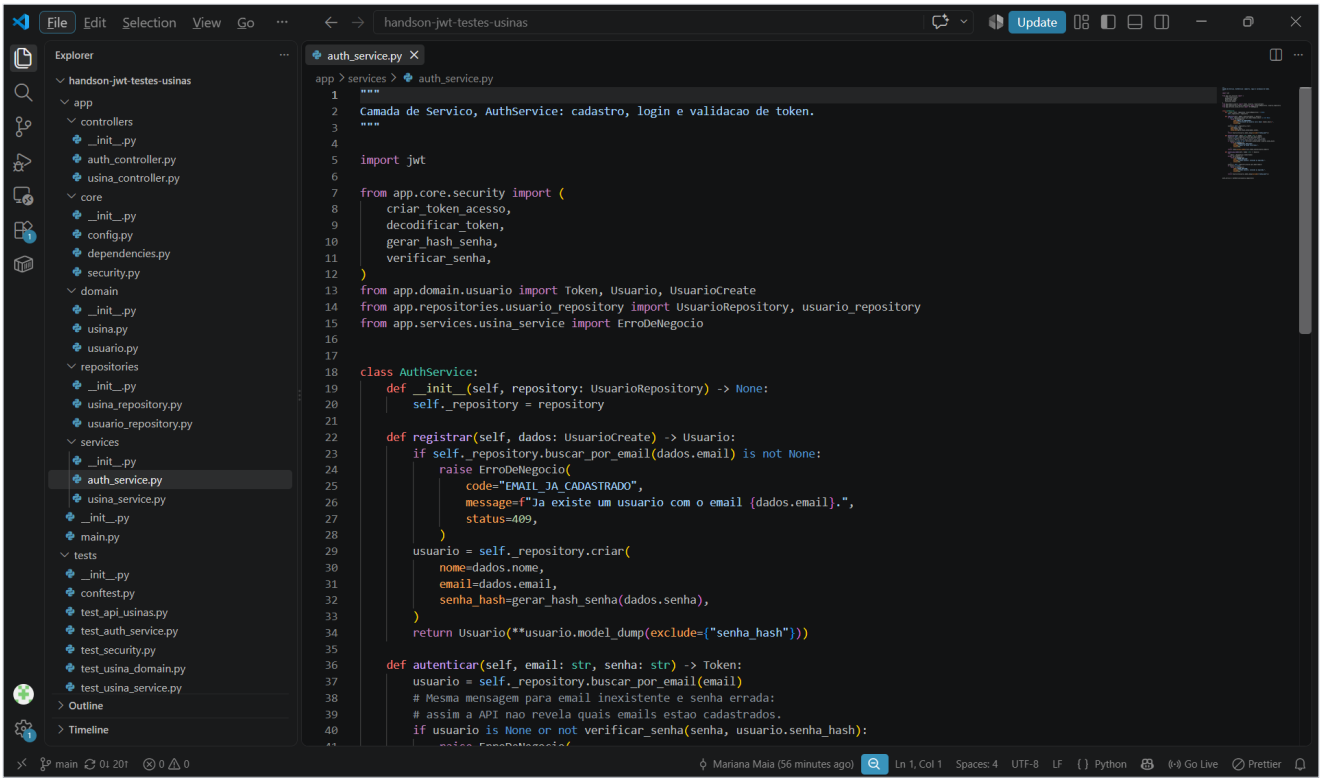
Passo 5. Domínio do usuário: app/domain/usuario.py

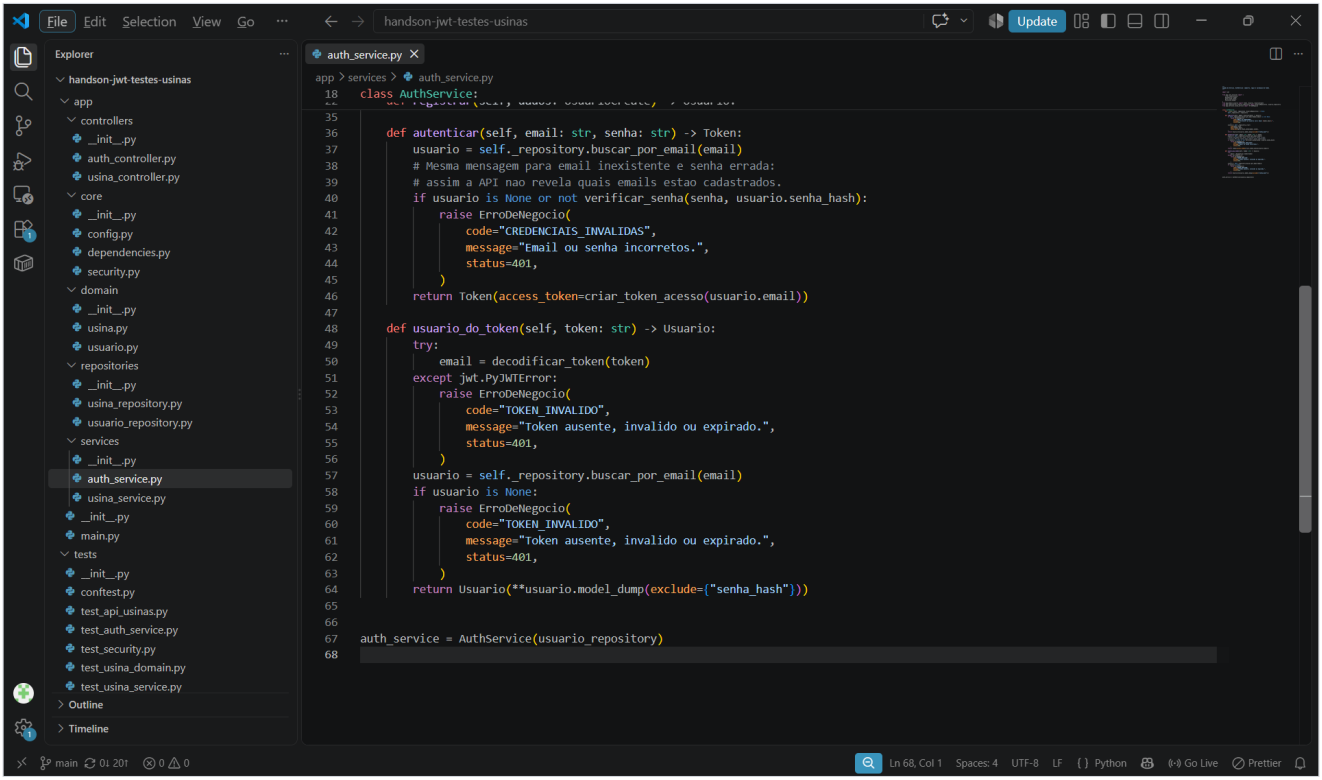


Passo 6. Persistência: app/repositories/usuario_repository.py

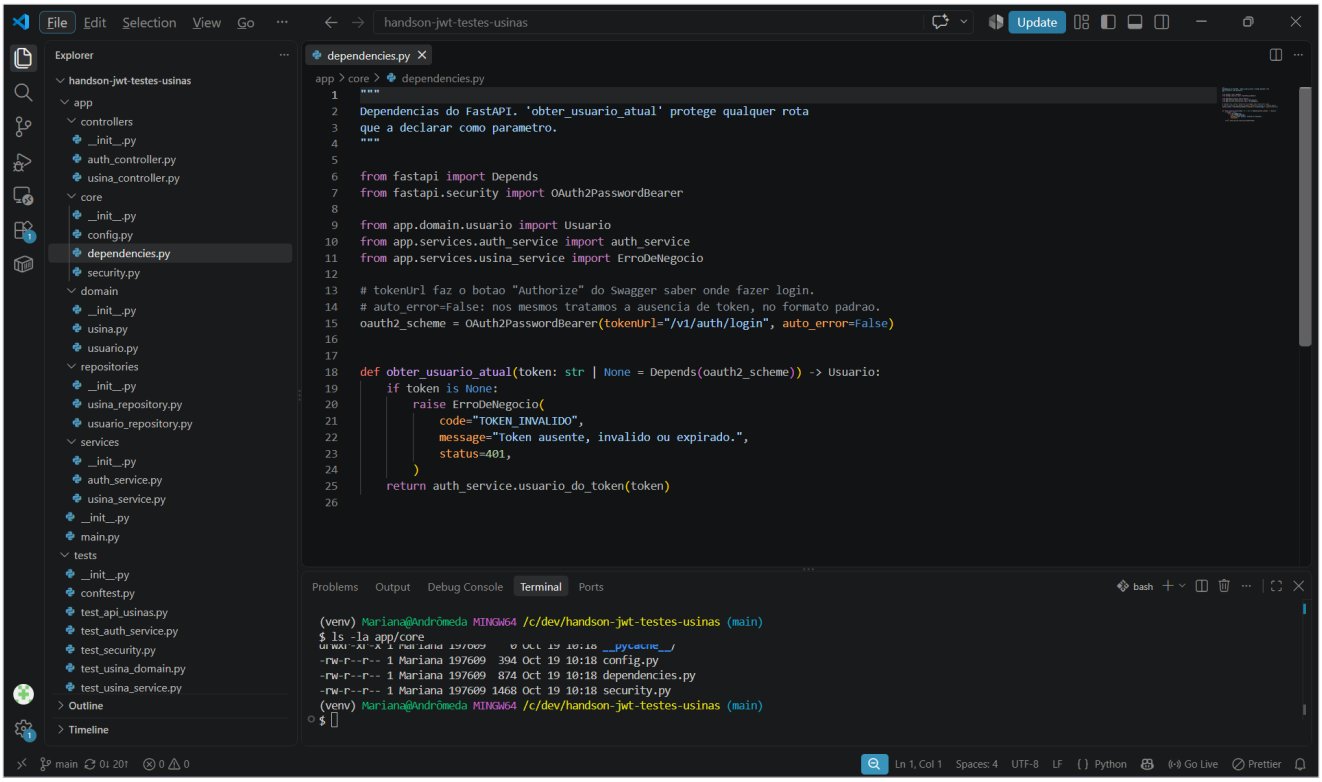


Passo 7. Serviço: app/services/auth_service.py

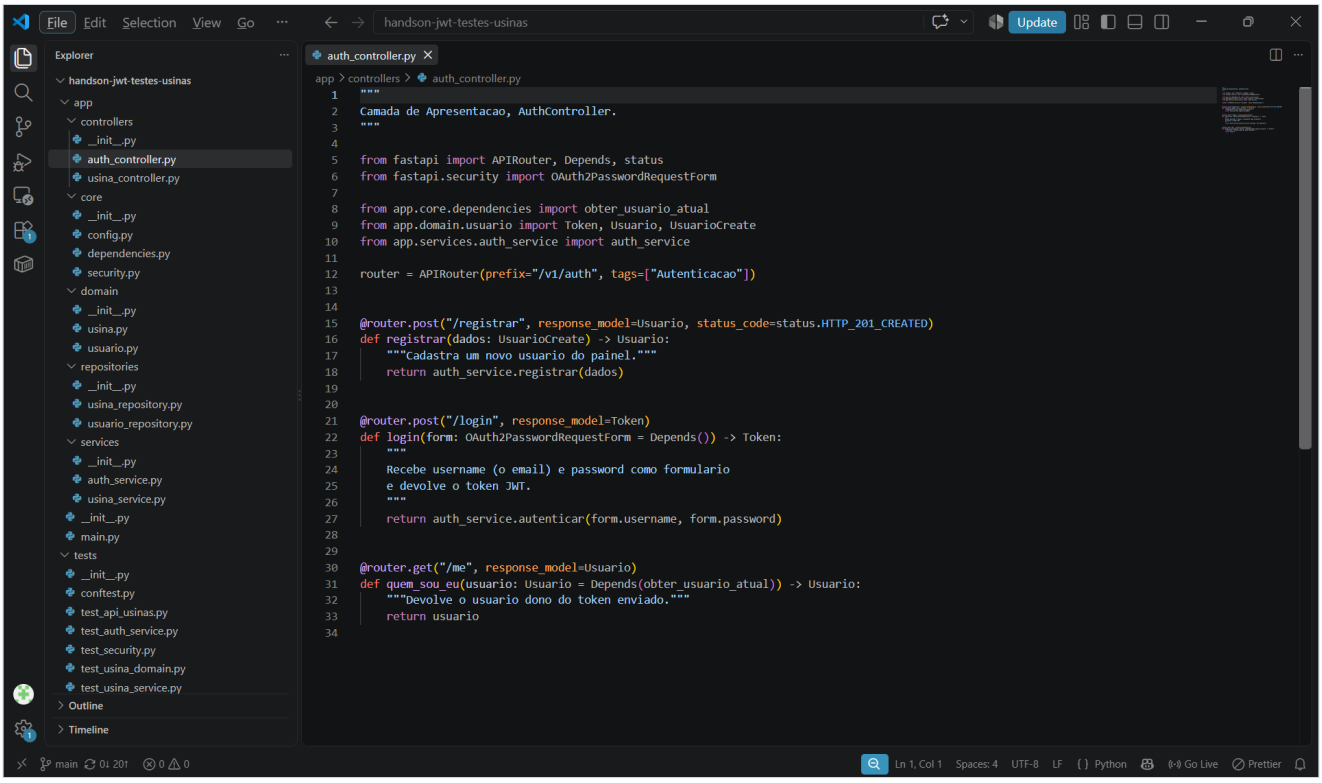




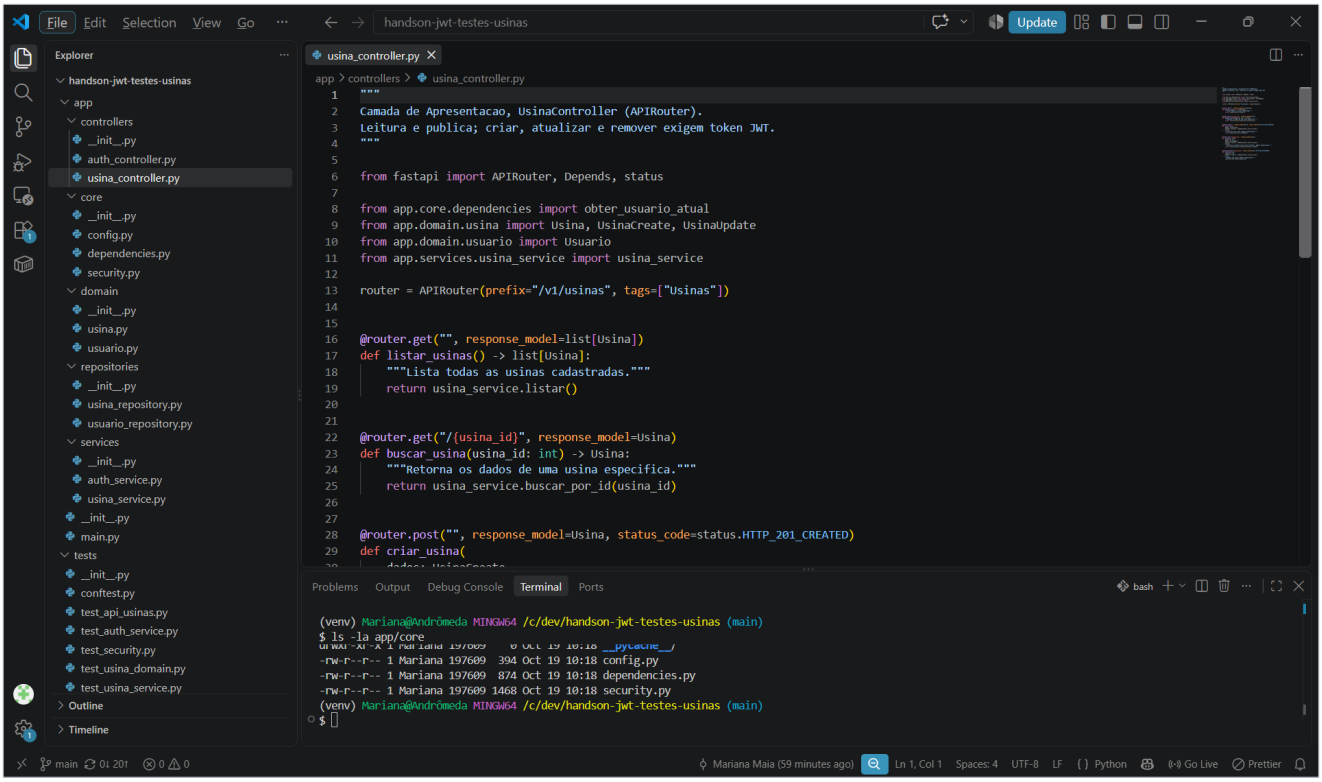
Passo 8. Dependência de proteção: app/core/dependencies.py

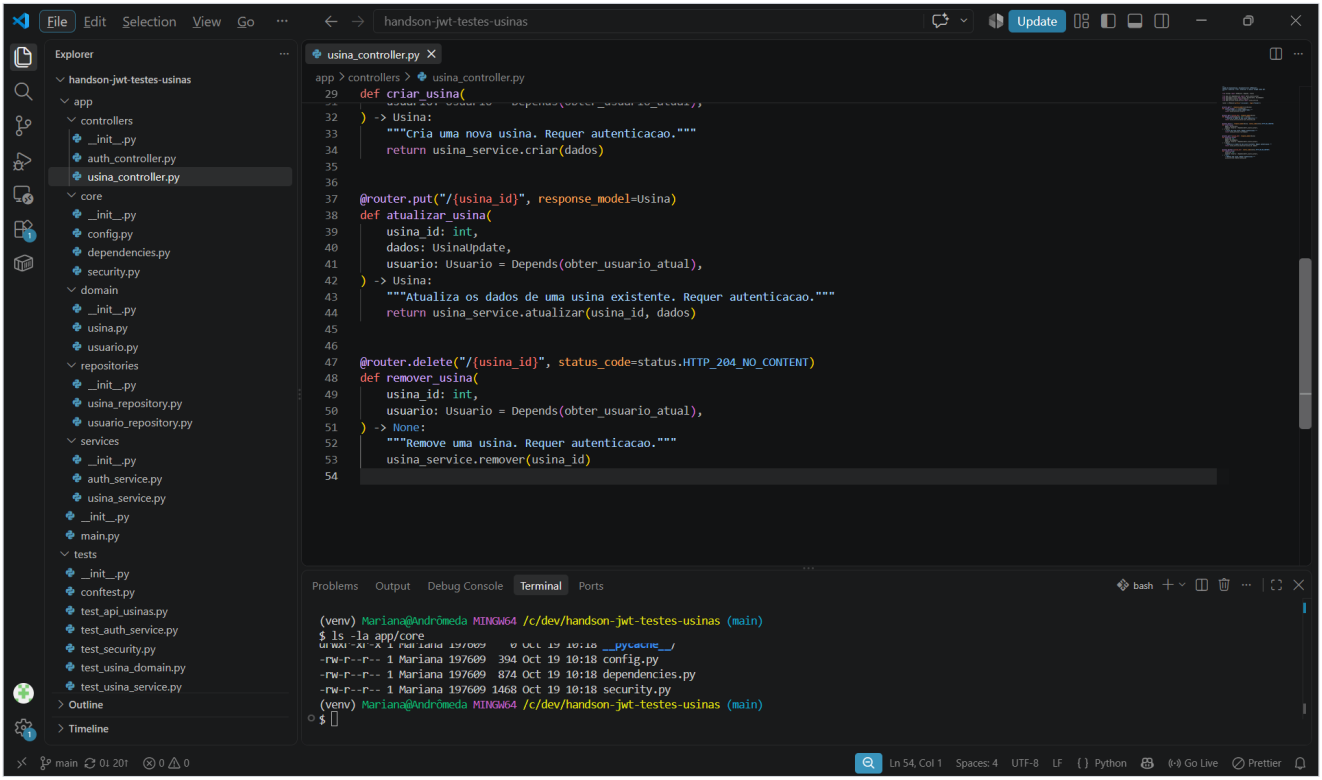


Passo 9. Controller de autenticação: app/controllers/auth_controller.py

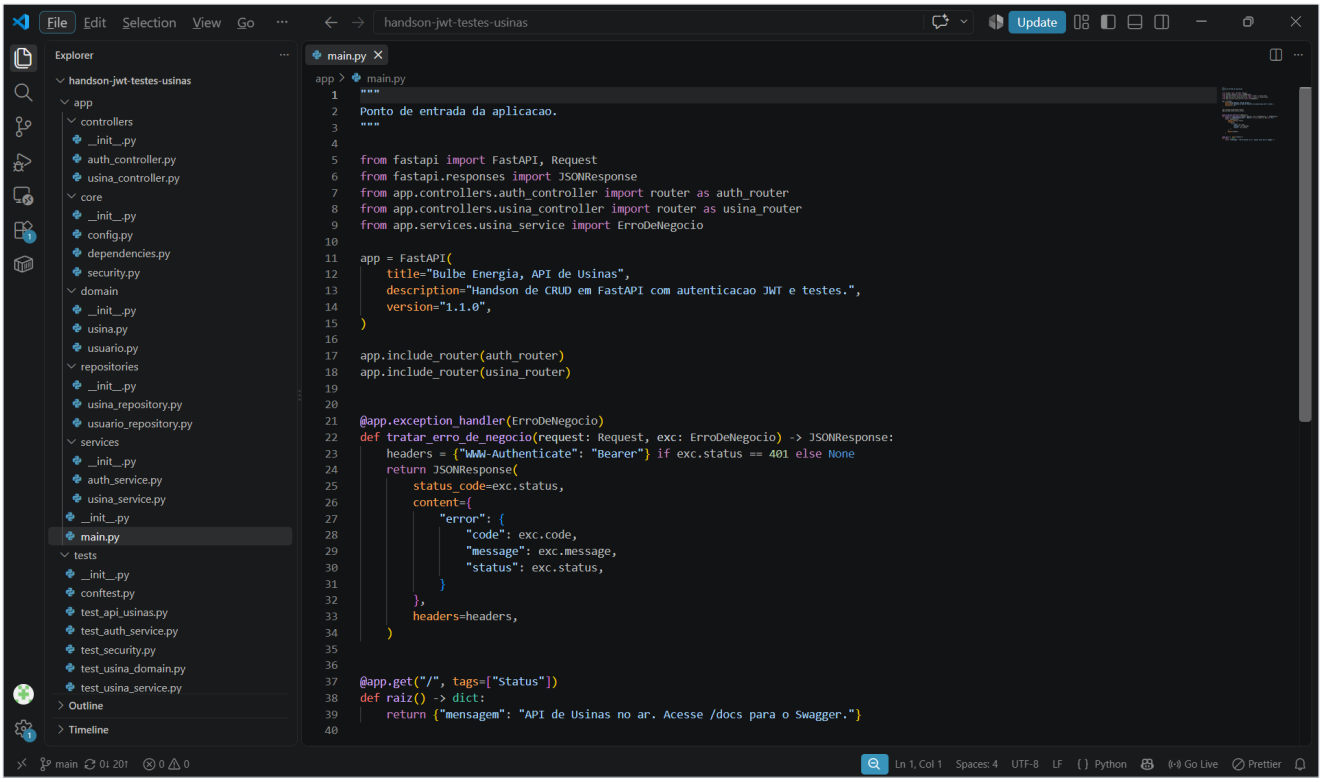


Passo 10. Proteger as rotas de Usina: app/controllers/usina_controller.py

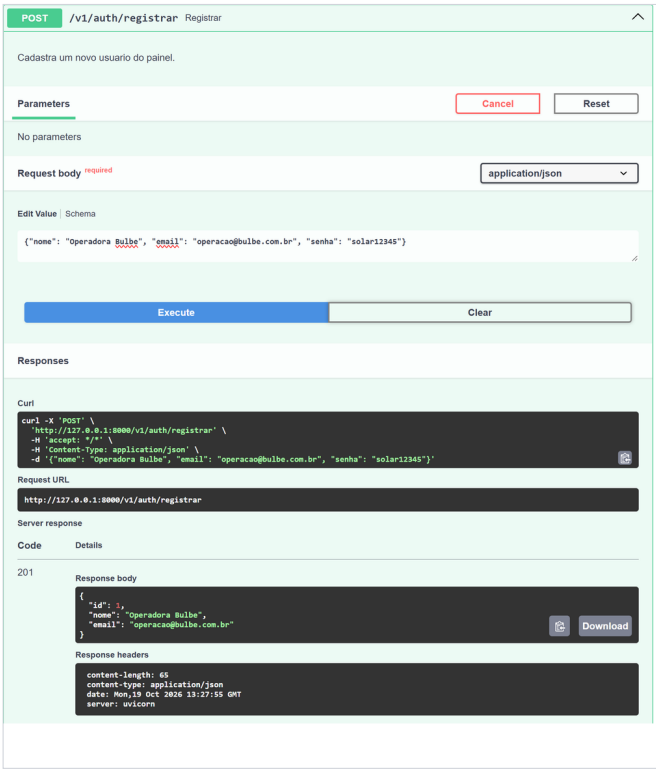


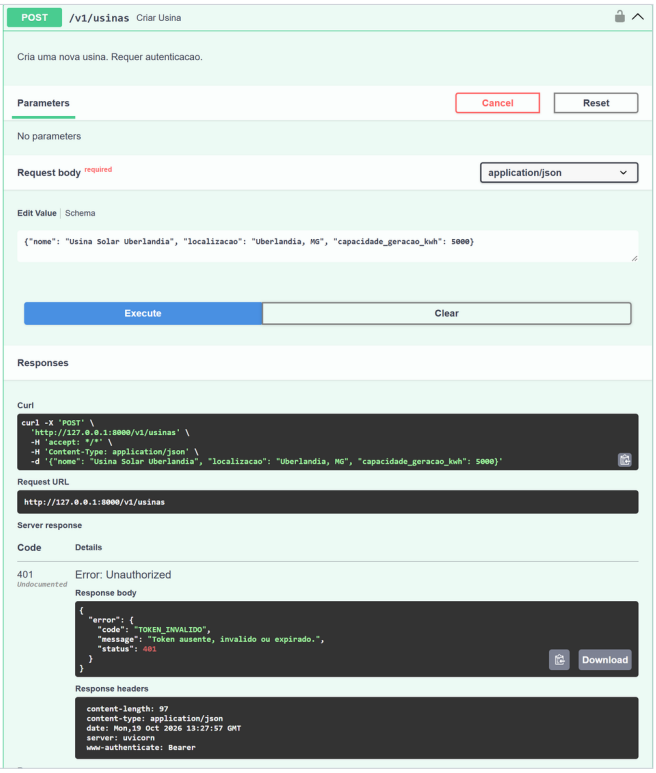


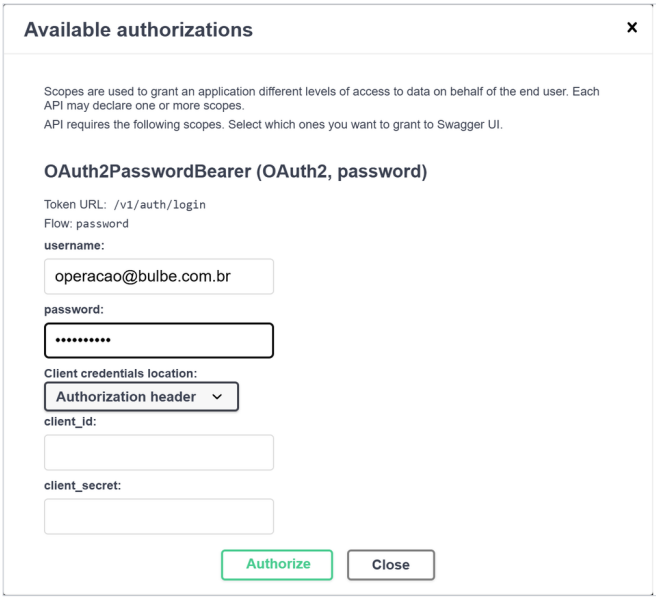
Passo 11. Registrar o router: app/main.py

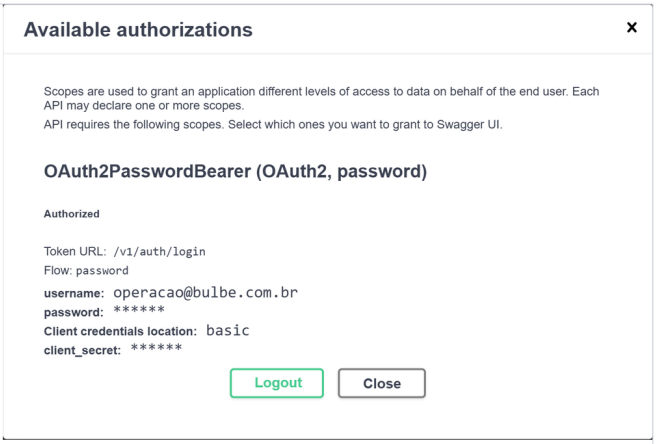


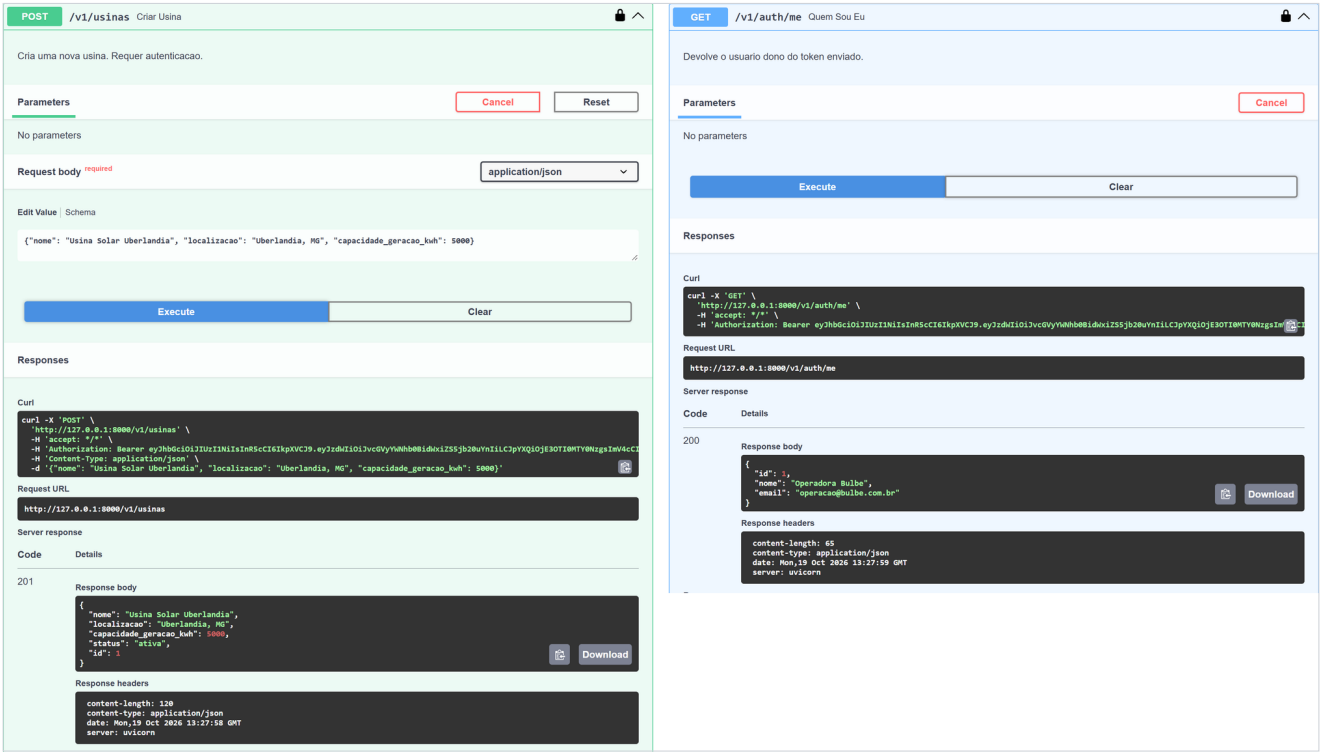
Passo 12. Testar pelo Swagger



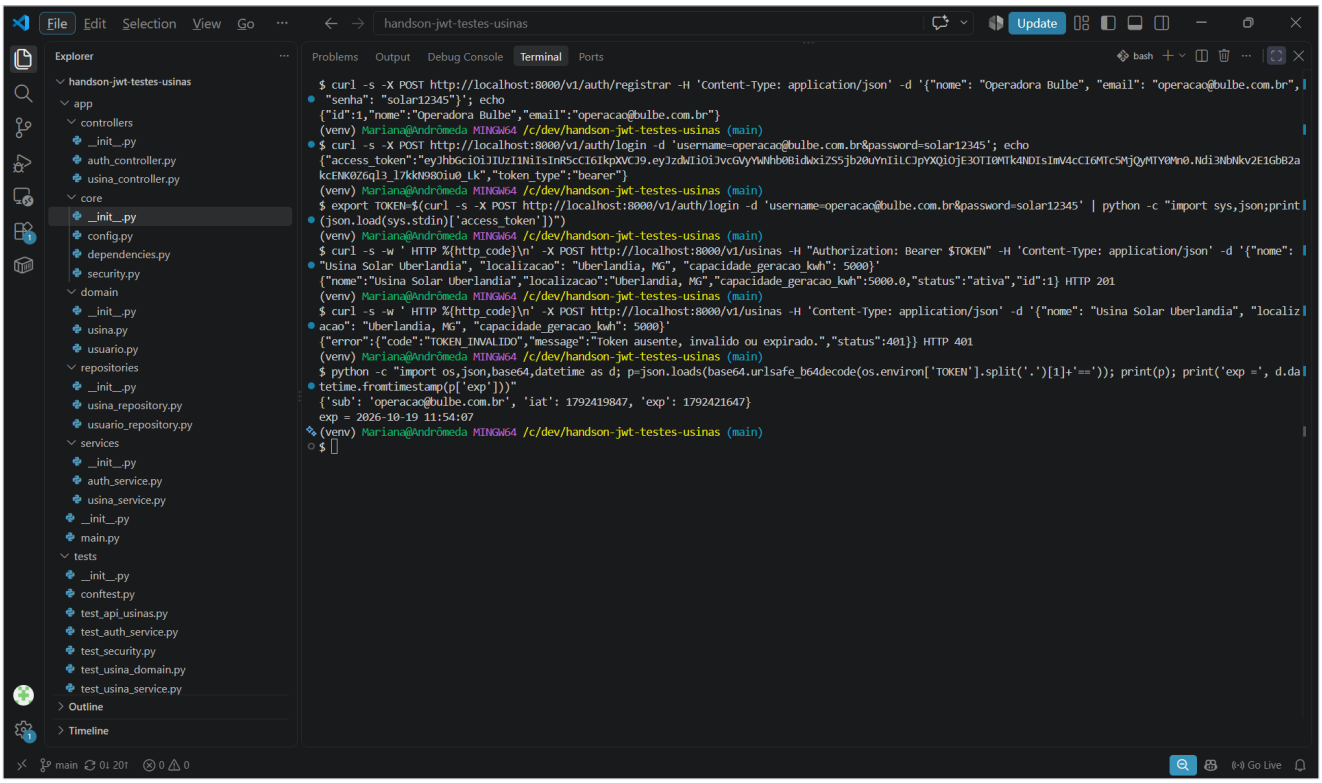






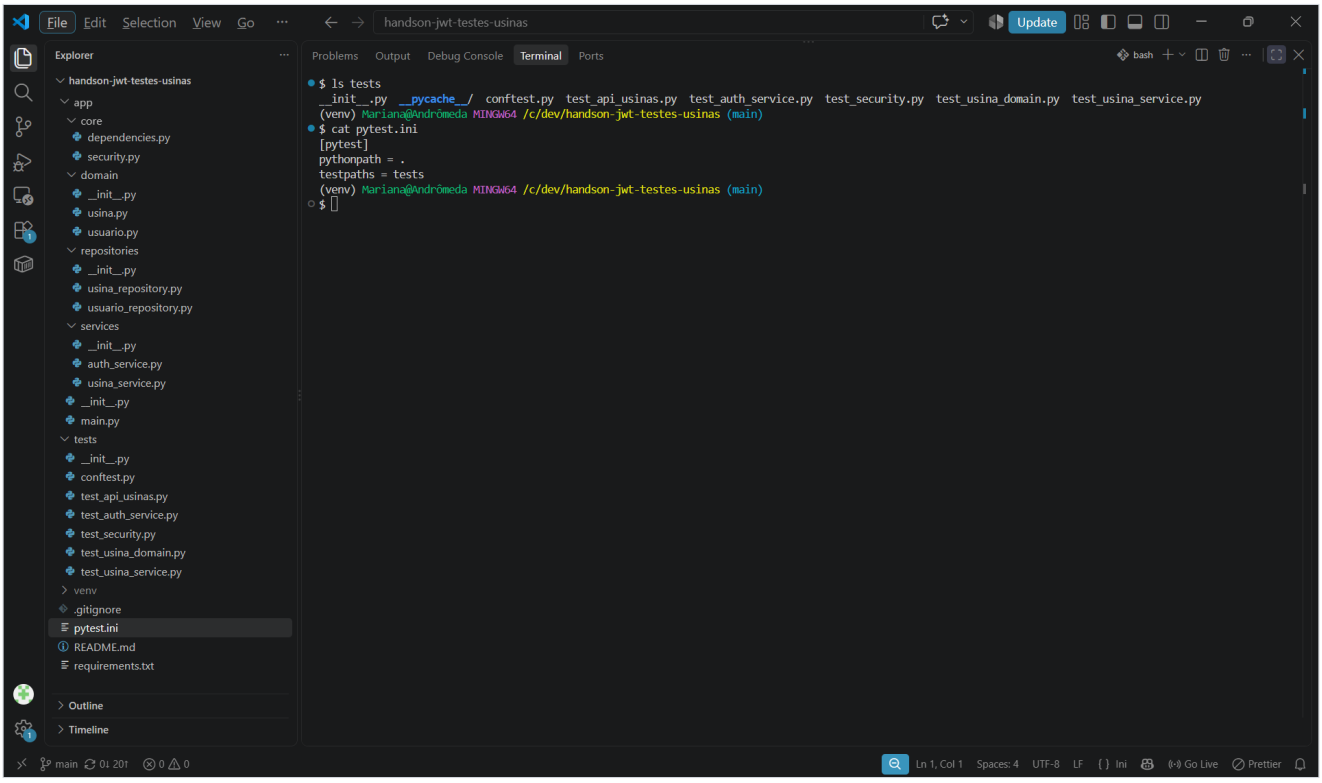


Passo 13. Testar pelo terminal

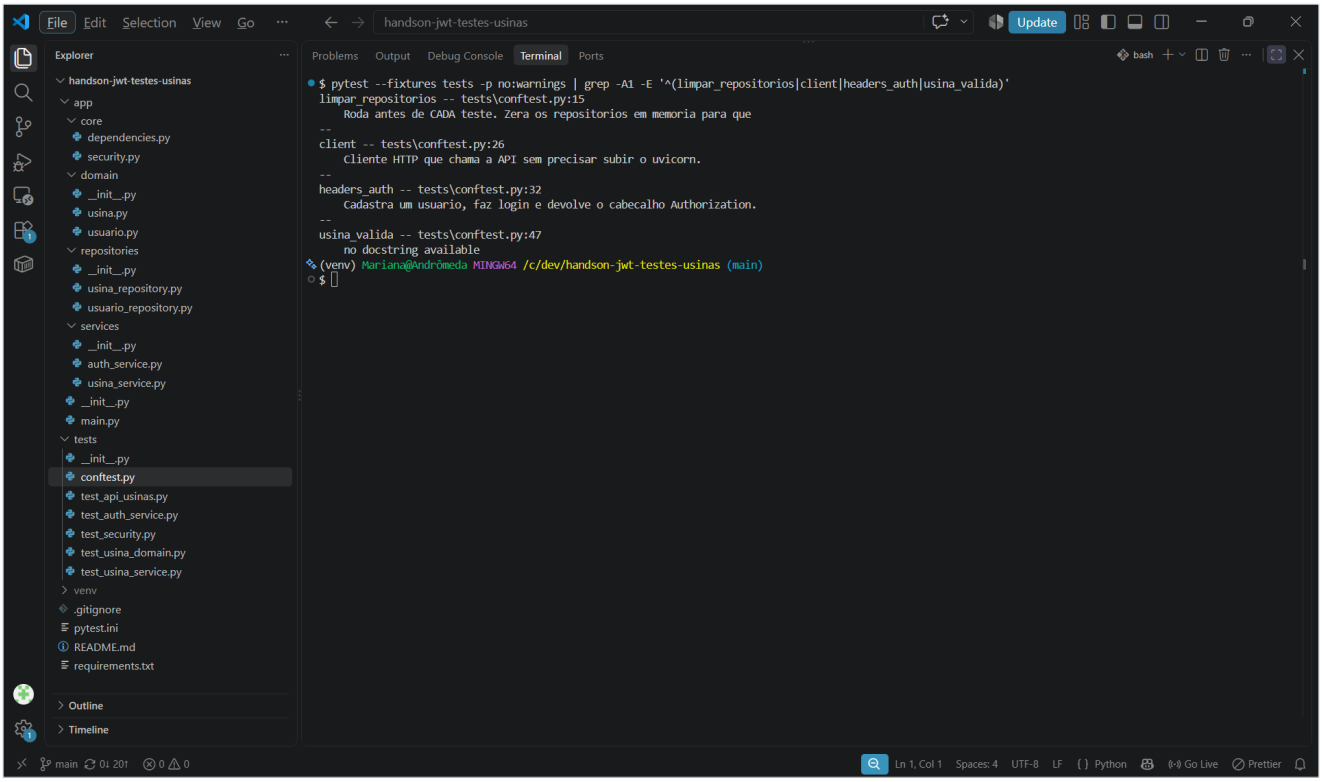


Parte 2. Testes unitários com pytest

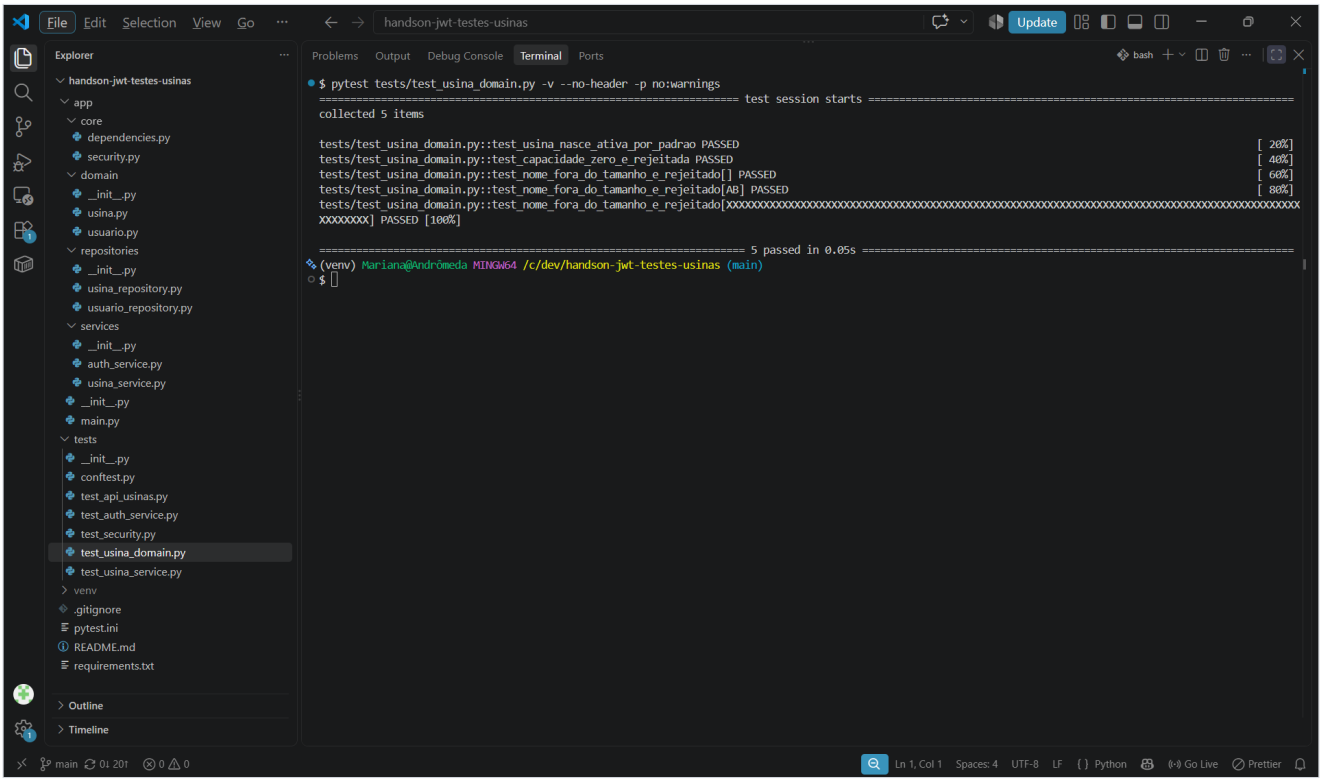
Passo 14. Criar a pasta de testes e configurar o pytest



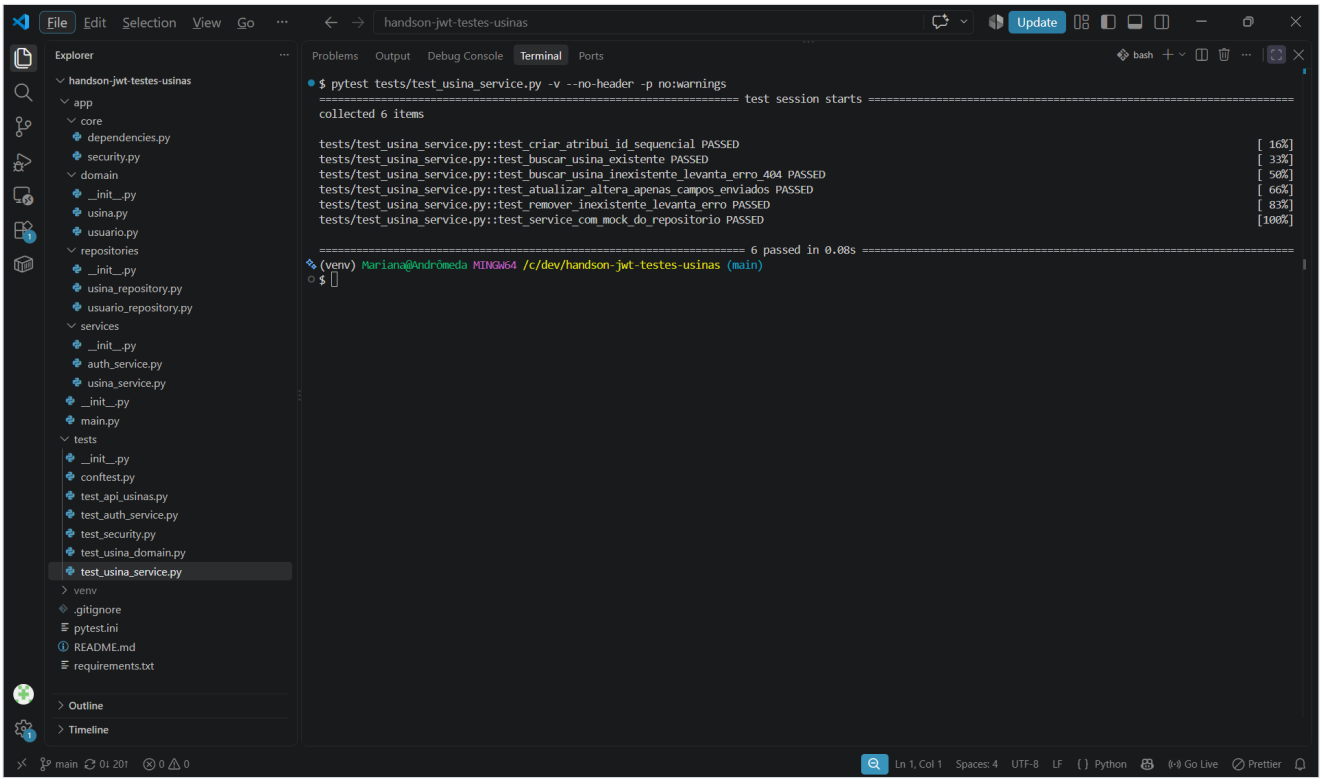
Passo 15. Fixtures compartilhadas: tests/conftest.py



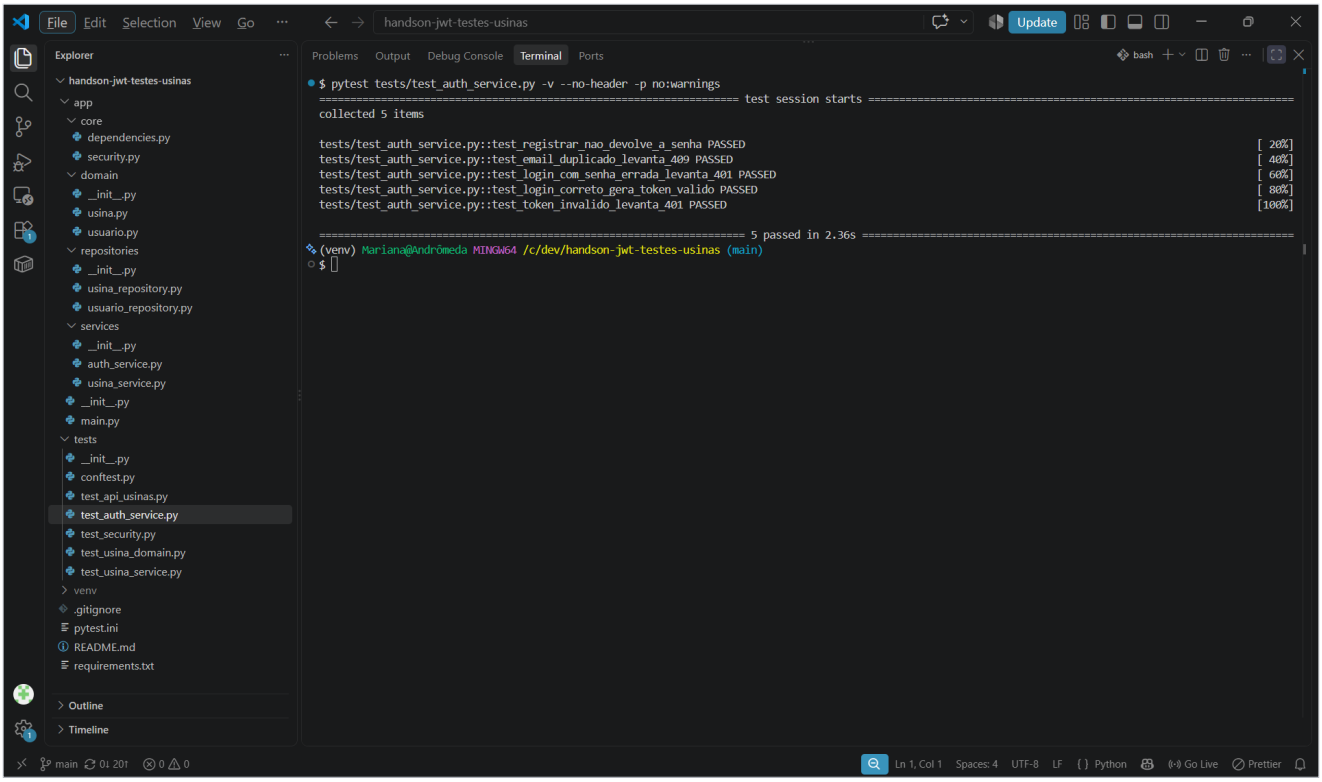
Passo 16. Testes do Domínio: tests/test_usina_domain.py


File Edit Selection View Go
handson-jwt-testes-usinas
Update
Explorer
handson-jwt-testes-usinas
app
core
dependencies.py
security.py
domain
__init__.py
usina.py
usuario.py
repositories
__init__.py
usina_repository.py
usuario_repository.py
services
__init__.py
auth_service.py
usina_service.py
__init__.py
main.py
tests
__init__.py
conftest.py
test_api_usinas.py
test_auth_service.py
test_security.py
test_usina_domain.py
test_usina_service.py
venv
.gitignore
pytest.ini
README.md
requirements.txt
Outline
Timeline
Problems Output Debug Console Terminal Ports
bash
$ pytest tests/test_usina_domain.py -v --no-header -p no:warnings
test session starts
collected 5 items
tests/test_usina_domain.py::test_usina_nasce_ativa_por_padrao PASSED [20%]
tests/test_usina_domain.py::test_capacidade_zero_e_rejeitada PASSED [40%]
tests/test_usina_domain.py::test_nome_fora_do_tamanho_e_rejeitado[] PASSED [60%]
tests/test_usina_domain.py::test_nome_fora_do_tamanho_e_rejeitado[AB] PASSED [80%]
tests/test_usina_domain.py::test_nome_fora_do_tamanho_e_rejeitado[XXXX

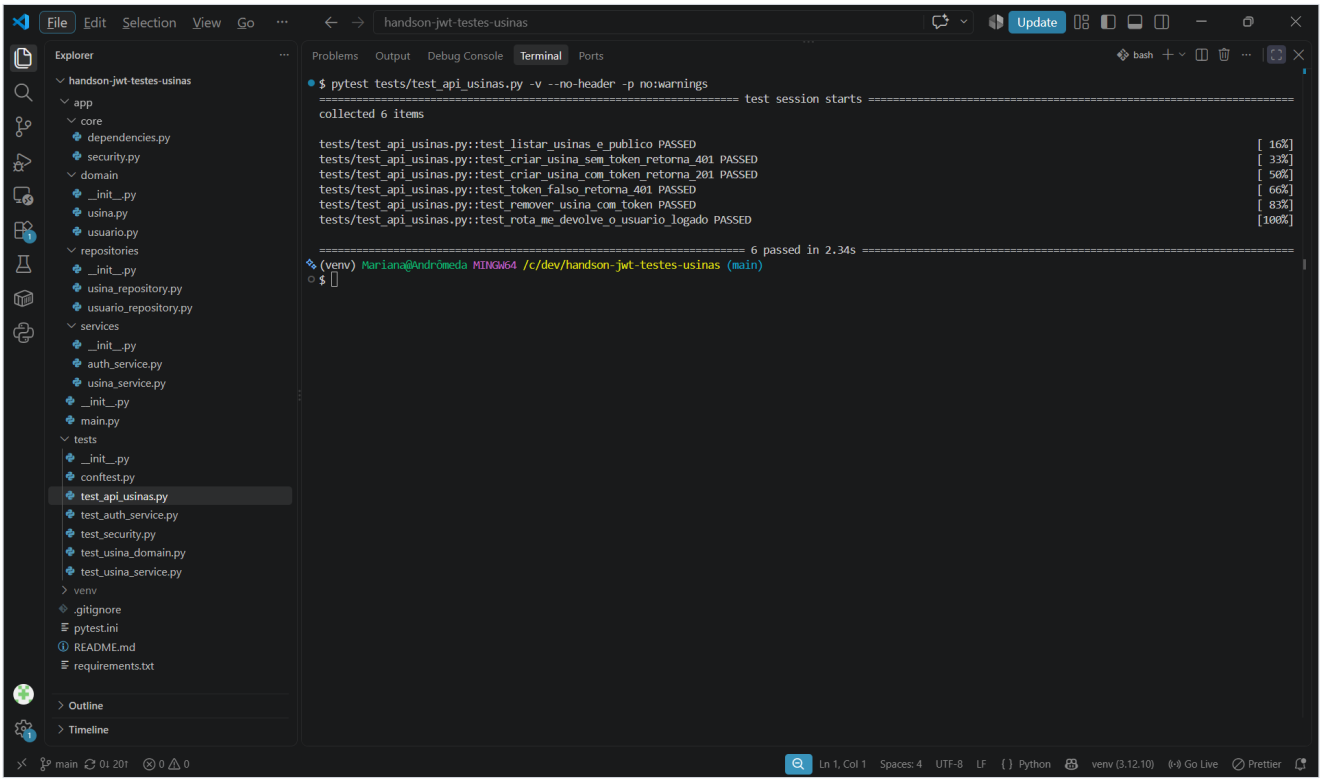
Passo 18. Testes do Service de Usinas: tests/test_usina_service.py



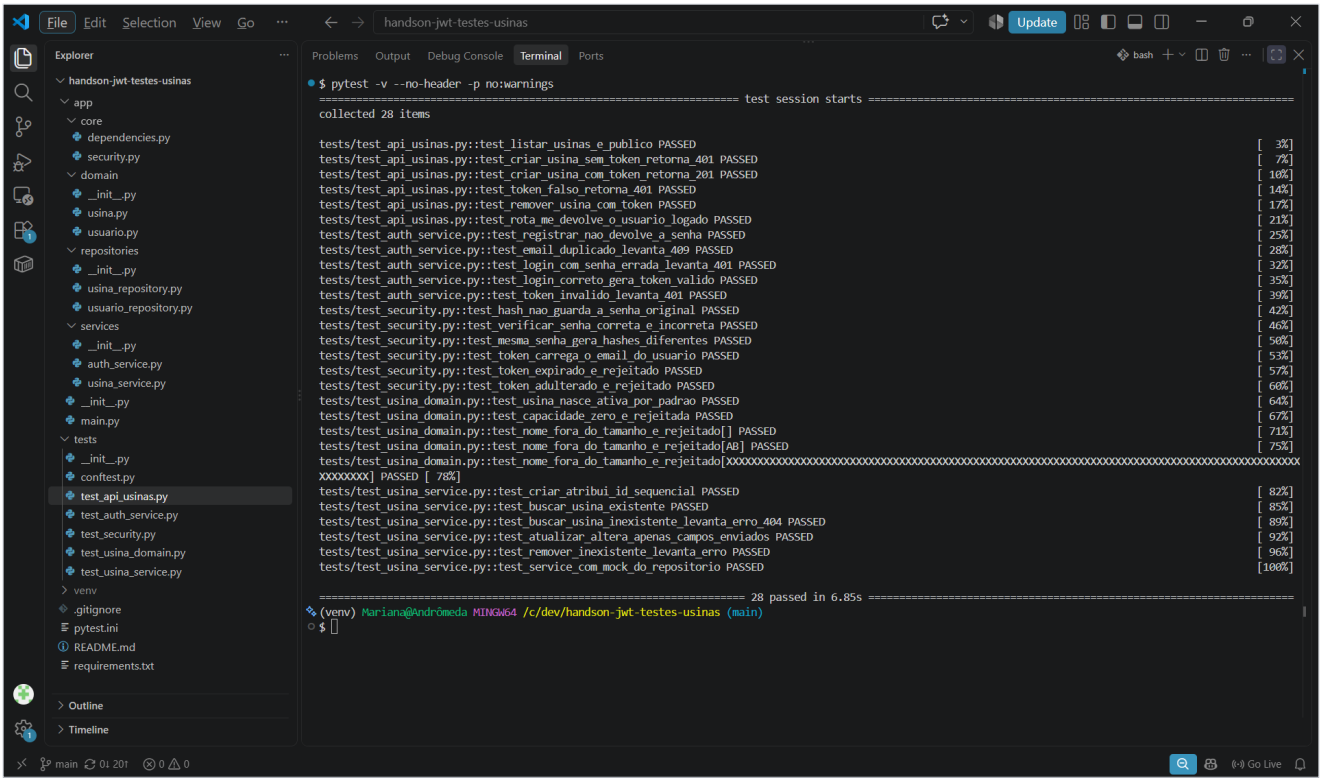
Passo 19. Testes do Service de Autenticação: tests/test_auth_service.py

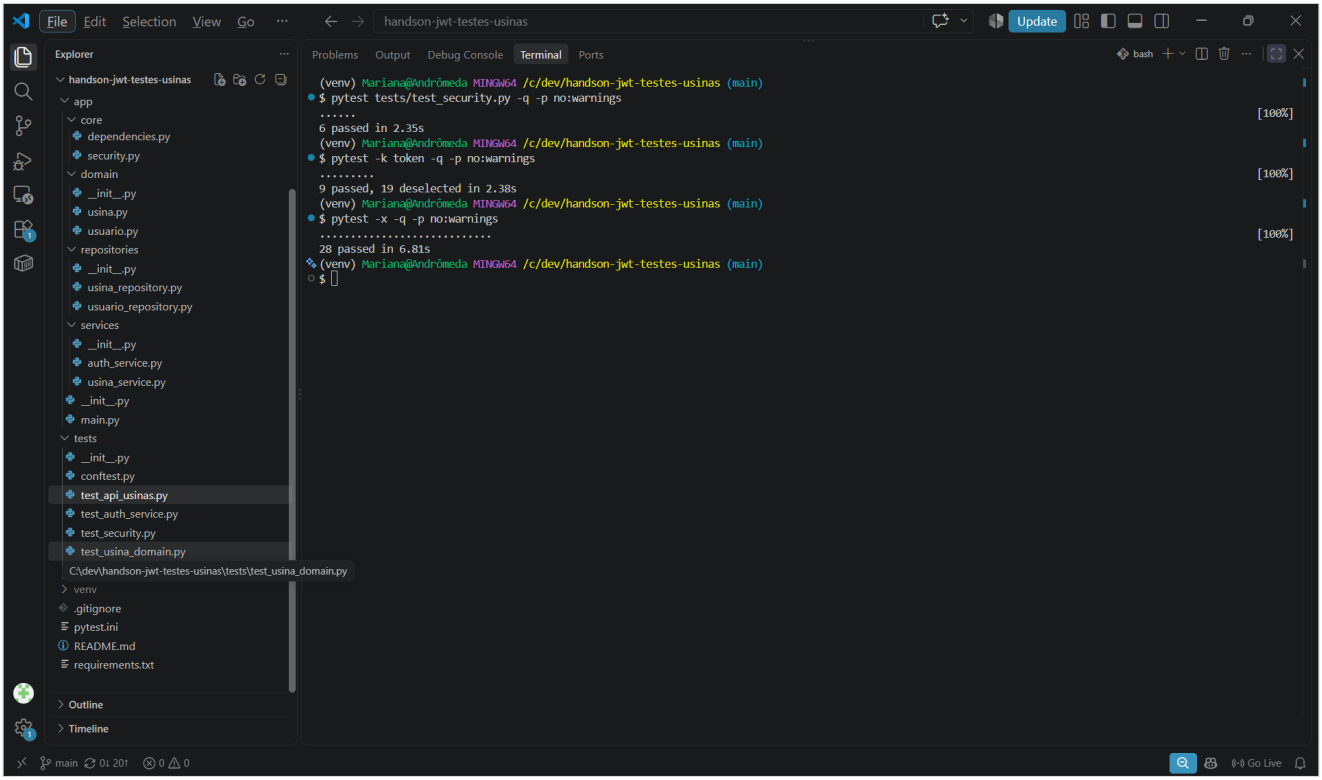


Passo 20. Testes da API: tests/test_api_usinas.py

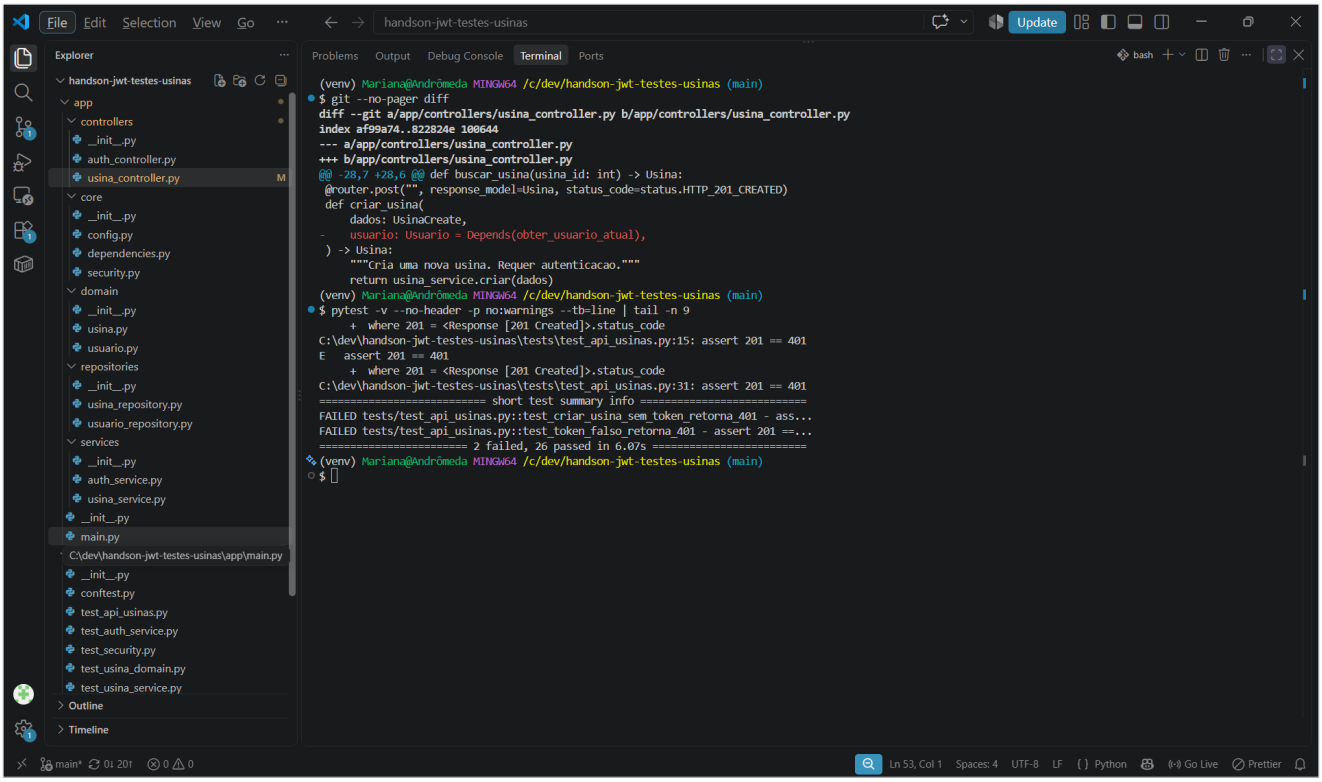


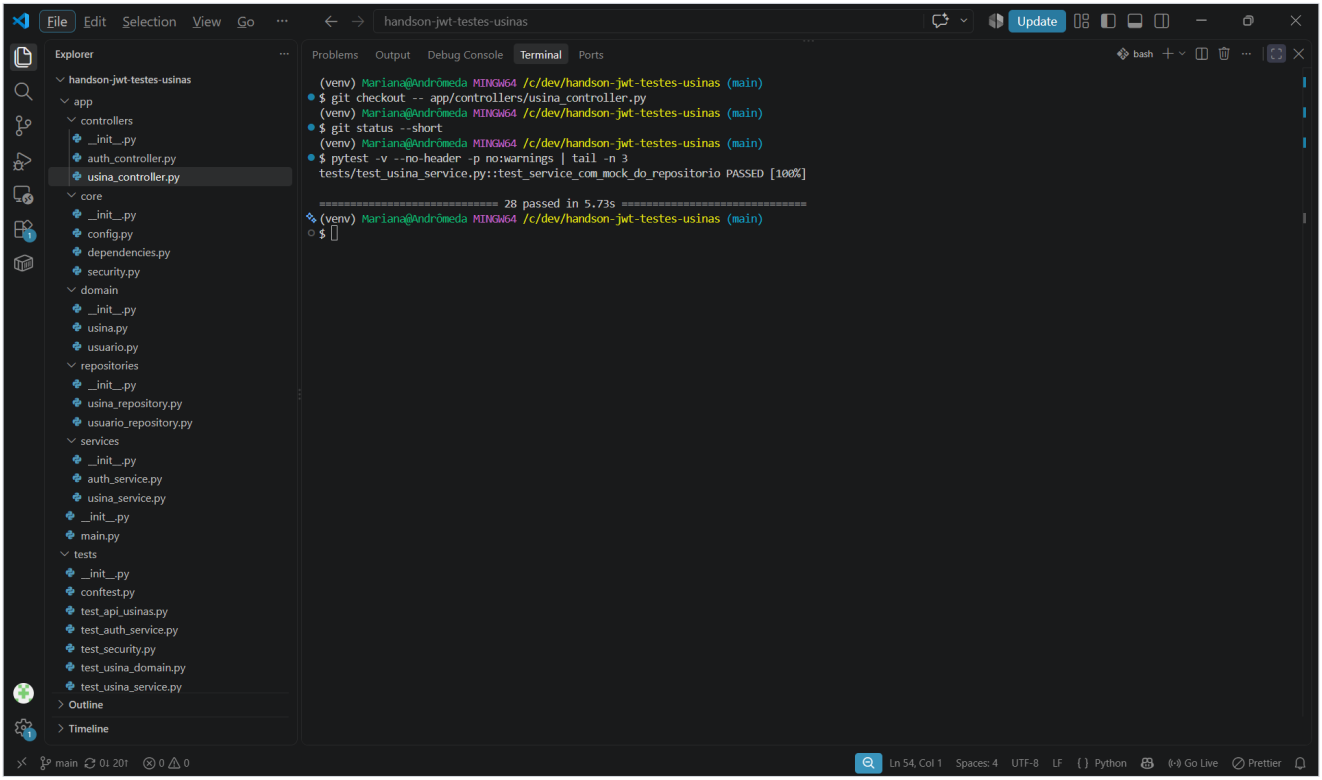
Passo 21. Rodar os testes


File Edit Selection View Go
handson-jwt-testes-usinas
Update
Explorer
handson-jwt-testes-usinas
app
core
dependencies.py
security.py
domain
__init__.py
usina.py
usuario.py
repositories
__init__.py
usina_repository.py
usuario_repository.py
services
__init__.py
auth_service.py
usina_service.py
__init__.py
main.py
tests
__init__.py
conftest.py
test_api_usinas.py
test_auth_service.py
test_security.py
test_usina_domain.py
test_usina_service.py
venv
.gitignore
pytest.ini
README.md
requirements.txt
Outline
Timeline
Problems Output Debug Console Terminal Ports
bash
$ pytest -v --no-header -p no:warnings
test session starts
collected 28 items
tests/test_api_usinas.py::test_listar_usinas_e_publico PASSED [3%]
tests/test_api_usinas.py::test_criar_usina_sem_token_retorna_401 PASSED [7%]
tests/test_api_usinas.py::test_criar_usina_com_token_retorna_201 PASSED [10%]
tests/test_api_usinas.py::test_token_falso_retorna_401 PASSED [14%]
tests/test_api_usinas.py::test_remover_usina_com_token PASSED [17%]
tests/test_api_usinas.py::test_rota_me_devolve_o_usuario_logado PASSED [21%]
tests/test_auth_service.py::test_registrar_nao_devolve_a_senha PASSED [25%]
tests/test_auth_service.py::test_email_duplicado_levanta_409 PASSED [28%]
tests/test_auth_service.py::test_login_com_senha_errada_levanta_401 PASSED [32%]
tests/test_auth_service.py::test_login_correto_gera_token_valido PASSED [35%]
tests/test_auth_service.py::test_token_invalido_levanta_401 PASSED [39%]
tests/test_security.py::test_hash_nao_guarda_a_senha_original PASSED [42%]
tests/test_security.py::test_verificar_senha_correta_e_incorreta PASSED [46%]
tests/test_security.py::test_mesma_senha_gera_hashes_diferentes PASSED [50%]
tests/test_security.py::test_token_carrega_o_email_do_usuario PASSED [53%]
tests/test_security.py::test_token_expirado_e_rejeitado PASSED [57%]
tests/test_security.py::test_token_adulterado_e_rejeitado PASSED [60%]
tests/test_usina_domain.py::test_usina_nasce_ativa_por_padrao PASSED [64%]
tests/test_usina_domain.py::test_capacidade_zero_e_rejeitada PASSED [67%]
tests/test_usina_domain.py::test_nome_fora_do_tamanho_e_rejeitado[] PASSED [71%]
tests/test_usina_domain.py::test_nome_fora_do_tamanho_e_rejeitado[AB] PASSED [75%]
tests/test_usina_domain.py::test_nome_fora_do_tamanho_e_rejeitado[XXXX



Passo 22. Ver um teste falhar (de propósito)




File Edit Selection View Go
handson-jwt-testes-usinas
Update
Explorer
handson-jwt-testes-usinas
app
controllers
__init__.py
auth_controller.py
usina_controller.py
core
__init__.py
config.py
dependencies.py
security.py
domain
__init__.py
usina.py
usuario.py
repositories
__init__.py
usina_repository.py
usuario_repository.py
services
__init__.py
auth_service.py
usina_service.py
__init__.py
main.py
tests
__init__.py
conftest.py
test_api_usinas.py
test_auth_service.py
test_security.py
test_usina_domain.py
test_usina_service.py
Outline
Timeline
Problems Output Debug Console Terminal Ports
bash
(venv) Mariana@Andrômeda MINGW64 /c/dev/handson-jwt-testes-usinas (main)
$ git checkout -- app/controllers/usina_controller.py
(venv) Mariana@Andrômeda MINGW64 /c/dev/handson-jwt-testes-usinas (main)
$ git status --short
(venv) Mariana@Andrômeda MINGW64 /c/dev/handson-jwt-testes-usinas (main)
$ pytest -v --no-header -p no:warnings | tail -n 3
tests/test_usina_service.py::test_service_com_mock_do_repositorio PASSED [100%]
========== 28 passed in 5.73s ==========
(venv) Mariana@Andrômeda MINGW64 /c/dev/handson-jwt-testes-usinas (main)
$
main
Ln 54, Col 1
Spaces: 4
UTF-8
LF
Python
Go Live
Prettier